

Graph Paths I

Time limit: 1.00 s

Memory limit: 512 MB

Consider a directed graph that has n nodes and m edges. Your task is to count the number of paths from node 1 to node n with exactly k edges.

Input

The first input line contains three integers n , m and k : the number of nodes and edges, and the length of the path. The nodes are numbered $1, 2, \dots, n$.

Then, there are m lines describing the edges. Each line contains two integers a and b : there is an edge from node a to node b .

Output

Print the number of paths modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 100$
- $1 \leq m \leq n(n - 1)$
- $1 \leq k \leq 10^9$
- $1 \leq a, b \leq n$

Example

Input	Output
3 4 8 1 2 2 3 3 1 3 2	2

Explanation: The paths are $1 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 3$ and $1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3$.